Blockchain for Impact Funding Evidence Map

Pilot Design Report, with Annex for the Full-Scope Evidence Map

To: GSR Foundation

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Partners named in this design: Mercy Corps Ventures (MCV), UNICEF Venture Fund, Filecoin Foundation, and GSR Foundation Blockchain for Good grantees

Version: 0.3. October 2026. Pilot sections are unchanged from v0.1, approved by GSR at Gate 0. Annex A adds participatory milestones (A7), data management, security and privacy (A10) and an itemised data request (A11).

Status and access: This report and all pilot outputs are held on a permission-restricted Google Drive. The design will be posted publicly once GSR approves it. All other outputs stay restricted until GSR approves the full-scope research, when a project website will be built.

Statement on AI use
This pilot uses large language models (LLMs) for design support, evidence collection, coding, analysis and drafting. One human researcher (Michael Cooper) is accountable for every judgement and approves each gate. Following the 2025 joint position statement of Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence, which supports the RAISE recommendations,[1] the pilot will report the models, versions and dates used; the purpose of each use; the prompts, raw outputs and code; how outputs were checked; and the limitations of AI use. This design report was drafted with LLM assistance and reviewed by the author.

[1] Flemyng, E. et al. (2025). Position statement on artificial intelligence (AI) use in evidence synthesis across Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence. Campbell Systematic Reviews. https://onlinelibrary.wiley.com/doi/full/10.1002/cl2.70074

1. Summary

The Blockchain for Impact Funding Evidence Map will show what blockchain-based impact projects aim to do, how they are delivered, how far they progress after the pilot stage, and how strong the evidence is behind their results.[2] Before committing to a full map of 800+ use cases, Metagov will run a four-week pilot on up to 30 publicly documented use cases. The pilot has two jobs: to test whether the method works, and to give GSR the information it needs to decide whether the full exercise is worth doing.

The design follows the evidence and gap map (EGM) standards of the Campbell Collaboration and 3ie as closely as the setting allows.[3][4] It differs from a conventional EGM in three disclosed ways, each justified below. First, cases are drawn from four partner portfolios rather than from a full literature search. Second, the map includes all levels of evidence, not just impact evaluations, because a map restricted to impact evaluations would be almost empty in this field. Third, two LLMs from different developers act as independent coders, with a third model suggesting resolutions to disagreements and the human researcher making every final decision.

Key design features:

- **Unit of analysis:** the use case, meaning one solution applied to one problem in one setting. Each use case is linked to multiple documents and has a lineage ID to prevent double counting across portfolios.
- **Sample:** up to 30 public use cases, about 7–8 from each of four portfolios, chosen by stratified purposive selection with random draws inside strata. Stalled and discontinued cases are deliberately included.
- **Framework:** a matrix of six use-case domains by six outcome domains, within Campbell's recommended size. It is supported by a five-level evidence scale with a crosswalk to the Nesta, Maryland and 3ie standards, plus separate codes for deployment status, blockchain's added value, independence of the evidence, and whether headline claims held up.
- **Reliability:** independent dual coding by two models, a calibration round, agreement statistics with confidence intervals, a 20% human audit of agreed codes, and human review of all disagreements, top-grade codes and low-confidence codes.
- **Oversight:** six approval gates (G0 to G5). GSR approves at G0 and G5 and confirms definitions at G1.
- **Decision:** the decision rule from the Purpose and Use Memo is applied unchanged. The full map proceeds if at least four of the five threshold metrics are met and the method fixes do not require redesigning the taxonomy or the evidence scale. GSR has final authority.[5]

2. Purpose, questions and scope

2.1 What the pilot must decide

The Evidence Map's primary use is targeting GSR's future Blockchain for Good funding. Its secondary uses are positioning GSR as a convenor of evidence and making the case to other donors.[6] The pilot serves those uses indirectly. It tells GSR whether a full map can be produced credibly, at reasonable cost, from the material available, and whether the answers are likely to change funding decisions.

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[2]Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[3]White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[4]Snilstveit, B. et al. (2016). Evidence & Gap Maps: A tool for promoting evidence informed policy and strategic research agendas. Journal of Clinical Epidemiology 79: 120-129. https://pubmed.ncbi.nlm.nih.gov/27387966/

[5]Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[6]Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

2.2 Pilot research questions

The pilot gives a first answer to all seven draft research questions in the Purpose and Use Memo. It also answers five method questions tied to the memo's pilot metrics.

#	Question	How the pilot answers it
RQ1–RQ2	What objectives, sectors, countries and implementation models do projects use, and which blockchain features do they rely on?	Descriptive coding of the use-case domain, objective, implementation model, blockchain function, funding model and country
RQ3	How far do projects advance after the pilot stage?	Deployment status coding and post-pilot status tracing (Section 6.3)
RQ4	What is the quality of the evidence: operational only, or social outcomes?	Five-level evidence scale applied to each outcome domain
RQ5	How well did projects achieve their stated objectives?	Objective-achievement code, linked to evidence level
RQ6	Does any evidence show blockchain added value over simpler alternatives?	Blockchain added-value code (0–3)
RQ7	Where is evidence weak or missing, and which gaps matter most for funding?	Gap typology (Section 9.1) and visual matrix
M1	Can the evidence scale be applied consistently?	Agreement between the two models (pilot metric 1)
M2	Is public material enough to grade a case?	Public-source coverage (metric 2)
M3	Do the categories cover real cases?	Taxonomy fit (metric 3)
M4	Can we find out what happened after each pilot?	Post-pilot status confirmed (metric 4)
M5	Is the full exercise affordable?	Effort per case (metric 5)

> **Justification.** Keeping the substantive questions identical to the memo preserves the agreed purpose, and lets the pilot test whether each question can be answered. Adding explicit method questions follows Campbell's guidance that the framework and coding tools should be piloted, in an iterative cycle of "revise, refine and define", before full coding.[7]

2.3 Changes from the Purpose and Use Memo

The design keeps the memo's purpose, four-week plan, metrics and decision rule. It makes four changes, which GSR is asked to confirm at Gate 0.

Memo text	This design	Reason
Pilot cases from MCV, the UNICEF Venture Fund and the Filecoin Foundation; GSR grantees added at full scale	Adds publicly documented GSR Blockchain for Good grantees as a fourth stratum, within the same 30-case limit	Tests the source-lineage and overlap rules before scale-up, because GSR funds MCV and UNICEF directly and co-funds a Filecoin cohort.[8][9] Uses public material only.

[7] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[8] GSR Foundation. How we fund. https://www.gsrfoundation.io/how-we-fund

[9] Filecoin Foundation (2025). Unlocking impact through infrastructure: Filecoin Foundation and the GSR Foundation work together to fund Filecoin use cases. https://fil.org/blog/unlocking-impact-through-infrastructure-filecoin-foundation-and-the-gsr-foundation-work-together-to-fund-filecoin-use-cases

Memo text	This design	Reason
Agreement measured with Cohen's kappa, threshold 0.7	Weighted kappa of 0.70 or more on the evidence grade stays the decision metric. Krippendorff's alpha, Gwet's AC1 and confidence intervals are also reported, with field-level quality bands.	Kappa can be misleading when one category dominates, which is likely here (most cases are expected to sit at the operational level). Reporting several statistics is standard practice and costs nothing extra.
Deployment status: pilot, extended pilot, scaled or discontinued	Adds "handed over" and "unknown"	The memo's Week 3 text already refers to hand-over. Without an "unknown" option, coders would be forced to guess.
Light taxonomy and five-level scale drafted in Week 2	Draft codebook v0.1 is included in this report (Annex B) and frozen as v1.0 at Gate 2	Leaves Week 2 for calibration rather than drafting, and lets GSR see the framework before the pilot starts

3. Standards and how they are adapted

3.1 Standards being pursued

This research is intended for audiences such as 3ie and the Campbell Collaboration, so it pursues their standards as far as possible. Use of blockchain for development and humanitarian outcomes is a new and fast-moving field. Most evidence is grey literature produced by implementers and funders, and much of it reports operational measures rather than social outcomes. The design treats this as a finding in its own right, not a reason to lower transparency. Where a standard cannot be met, the design says so, explains why, and states what replaces it.

- **Campbell EGM guidance** for framework design, coding, analysis and reporting.[10]
- **3ie EGM methods** for the intervention-by-outcome framework, stakeholder consultation and gap typology.[11][12]
- **ROSES for systematic maps** and a PRISMA-style flow diagram for transparent reporting of the case flow.[13]
- **Cochrane rapid review guidance** as the reference for justified, disclosed methodological shortcuts.[14]
- **The 2025 joint position statement on AI in evidence synthesis and the RAISE recommendations** for responsible, reported LLM use.[15]
- **Davies (2013) evaluability assessment** for the forward-looking layer in the full scope.[16]

[10] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[11] Snilstveit, B. et al. (2016). Evidence & Gap Maps: A tool for promoting evidence informed policy and strategic research agendas. Journal of Clinical Epidemiology 79: 120-129. https://pubmed.ncbi.nlm.nih.gov/27387966/

[12] Snilstveit, B. et al. (2017). 3ie evidence gap maps: a starting point for strategic evidence production and use. 3ie Working Paper 28. https://www.3ieimpact.org/sites/default/files/2019-01/wp28-egm.pdf

[13] ROSES for systematic map reports. https://www.roses-reporting.com/systematic-map-reports

[14] Cochrane Rapid Reviews Methods Group. Cochrane rapid review methods guidance (Garritty et al., BMJ 2024). https://methods.cochrane.org/rapidreviews/cochrane-rr-methods

[15] Flemyng, E. et al. (2025). Position statement on artificial intelligence (AI) use in evidence synthesis across Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence. Campbell Systematic Reviews. https://onlinelibrary.wiley.com/doi/full/10.1002/cl2.70074

[16] Davies, R. (2013). Planning evaluability assessments: a synthesis of the literature with recommendations. DFID Working Paper 40. https://www.gov.uk/government/publications/planning-evaluability-assessments

3.2 Alignment with each standard

Element	Conventional standard	Pilot approach	Justification
Protocol	Campbell expects a protocol (title registration, PICOS, framework, search, coding, analysis), with the search, screening and coding form piloted first.[17]	This report is the protocol. It is not registered, but is posted publicly once GSR approves it.	Public posting before any coding gives the same protection against changing the method after seeing results. Formal registration is better suited to the full scope.
Search	Several databases with documented search strings, double screening and a PRISMA diagram.[18]	A census of each partner's public portfolio, followed by targeted searches for each case, all logged.	The memo defines the pilot as a test of partner portfolios.[19] Database searches would return little on these cases. The full scope adds a systematic grey-literature search (Annex A).
Evidence included	3ie maps include impact evaluations and systematic reviews only.[20]	All evidence levels, from claim-only to counterfactual impact	Restricting to impact evaluations would leave the map nearly empty. Grading everything shows where evidence is thin, which is the point for GSR.
Screening and coding	Two independent reviewers, with disagreements resolved by a third.[21]	Two LLMs from different developers code independently. A third model suggests resolutions, and the human researcher decides and audits.	With one human, two independent model coders are the closest available equivalent of dual review. The human audit tests whether both models are wrong in the same way.
Critical appraisal	Recommended but not mandatory for Campbell maps.[22] 3ie appraises systematic reviews only.[23]	Evidence level, independence of the evidence, and a claims check	Formal risk-of-bias tools assume study designs that are rare here. Light appraisal of L4–L5 studies is added in the full scope.
Framework	4–6 rows and 4–6 columns, comprehensive and mutually exclusive, developed with stakeholders.[24]	Six by six, validated with GSR at G0 and tested in calibration	Directly follows the guidance
Reporting	Descriptive report, ROSES or PRISMA flow, and a plain-language summary.[25][26]	Same, plus an AI-use disclosure	Directly follows the guidance
Updating	Annual updating recommended.[27]	A reusable data structure so grantee reports can extend the map	Supports the memo's commitment to a map that can be updated[28]

[17] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[18] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[19] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[20] Snilstveit, B. et al. (2017). 3ie evidence gap maps: a starting point for strategic evidence production and use. 3ie Working Paper 28. https://www.3ieimpact.org/sites/default/files/2019-01/wp28-egm.pdf

[21] 3ie (2025). Private sector engagement evidence gap map protocol. https://3ieimpact.org/sites/default/files/2025-05/PSE-EGM-Protocol.pdf

[22] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[24] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[23] Snilstveit, B. et al. (2017). 3ie evidence gap maps: a starting point for strategic evidence production and use. 3ie Working Paper 28. https://www.3ieimpact.org/sites/default/files/2019-01/wp28-egm.pdf

[25] Campbell Collaboration (2020). Editorial: Evidence and gap maps. https://www.campbellcollaboration.org/2020/03/editorial-journal-issue1-2020/

[26] ROSES for systematic map reports. https://www.roses-reporting.com/systematic-map-reports

3.3 Adapting to thin, novel evidence

The following adaptations respond to the expected low quality and volume of data. Each one is built into the pilot, and their usefulness is reviewed at G5.

1. **Grade what exists rather than exclude it.** An inclusive five-level scale replaces impact-evaluation-only inclusion, so thin evidence becomes visible as a gap rather than disappearing.
2. **Grade each outcome separately.** A use case can have L3 evidence on cost and L1 on wellbeing. A single grade per case would hide this.
3. **Separate "not reported" from "not applicable".** Every field allows both codes, so missing evidence is never confused with evidence of no effect.
4. **Code who produced the evidence.** The codes are self-reported by the implementer, partner or funder, independent, or peer-reviewed. Most material will come from parties with an interest in positive results, so this filter lets readers see the map with self-reported evidence removed.
5. **Check claims against what happened next.** Headline claims, such as a 75% cost reduction, are logged in a claims register and checked against later sources and post-pilot status. This tests the memo's question of whether pilots are inflating utility.[29]
6. **Trace post-pilot status from digital signals.** Signals include whether the product is live, the latest public activity, follow-on funding, hand-over announcements and, where relevant, on-chain or code-repository activity. Each is coded with a confidence level.
7. **Treat on-chain data as operational evidence only.** Transaction records can verify usage and speed, but not social outcomes. The codebook caps on-chain evidence at L2 unless it is linked to outcome data.
8. **Code blockchain's added value separately from programme success.** A cash transfer can succeed without the ledger contributing anything. A separate 0–3 code asks whether any evidence compares the blockchain approach with a non-blockchain alternative.
9. **Deliberately sample failure.** Positive results dominate public communications. The sampling plan requires stalled or discontinued cases in the sample (Section 5.3).
10. **Use lineage IDs.** One use case can appear in several portfolios. A lineage ID links all records of the same deployment so the map does not count it twice.
11. **Report uncertainty.** Agreement statistics are reported with confidence intervals, and every code carries a confidence rating and a supporting quote.

4. Design overview

4.1 Unit of analysis and lineage

The unit of analysis is the use case: one solution applied to one problem in one setting, as defined in the memo.[30] A use case can be linked to many documents. The same solution deployed in two countries counts as two use cases, linked by a shared solution ID. If two portfolios support the same deployment, it is one use case with one lineage ID and a list of all funders, and it is assigned to a primary portfolio for sampling. For example, Akashic appears both in the Filecoin Foundation and GSR co-funded cohort and among GSR's Blockchain for Good grantees.[31][32]

[27] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[28] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[29] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[30] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

> **Justification.** Use cases match how GSR makes funding decisions and how partner portfolios are organised. Linking documents to use cases, rather than treating each document as a study, avoids counting several press releases about one pilot as several pieces of evidence.

4.2 Conceptual framework

Each use case is coded along a simple results chain: blockchain function → implementation model → outputs and usage → operational outcomes (cost, speed, transparency) → social outcomes (resilience, access, sector outcomes) → impact against a counterfactual. Coding also follows a deployment path: pilot → extended pilot → scaled, handed over or discontinued. The evidence scale shows how far along the results chain the evidence reaches. Deployment status shows how far along the deployment path the use case actually travelled. The pilot compares the two (metric 6).

4.3 Roles

Role	Who	Responsibilities
Principal investigator and sole human reviewer	Michael Cooper	Approves all gates, makes final adjudication decisions, carries out audits, and is accountable for every grade and conclusion
Orchestrator	LLM research assistant (Anthropic Claude family, as used for this design)	Builds the sampling frame, collects documents, assembles evidence packets, runs the coding pipeline, computes statistics, and drafts outputs. Does not code cases.
Coder A	Frontier LLM from a second developer (proposed: OpenAI GPT family)	Codes every case independently from the evidence packet and codebook
Coder B	Frontier LLM from a third developer (proposed: Google Gemini family)	Codes every case independently, with no access to Coder A's output
Adjudication assistant	Orchestrator model, run in a separate session	For each disagreement, reviews both codes and their quotes, proposes a resolution with reasons, and flags codebook ambiguity. Michael decides.
Partners	MCV, UNICEF Venture Fund, Filecoin Foundation, GSR grantees	In the pilot, none: public sources only. In the full scope, data providers under data-sharing agreements and factual reviewers before publication.
GSR Foundation	Funder	Approves the design (G0), confirms definitions (G1), and makes the scale-up decision (G5)

> **Justification.** Coders from different developers reduce the risk of correlated errors from shared training data. Keeping the orchestrator out of coding stops the model that assembled the evidence from grading it. Exact model versions are fixed at G2 and recorded in a model register.

[31] Filecoin Foundation (2025). Unlocking impact through infrastructure: Filecoin Foundation and the GSR Foundation work together to fund Filecoin use cases. https://fil.org/blog/unlocking-impact-through-infrastructure-filecoin-foundation-and-the-gsr-foundation-work-together-to-fund-filecoin-use-cases

[32] GSR Foundation. Home page (grantee listings). https://www.gsrfoundation.io/

5. Sampling plan

5.1 Eligibility criteria

Dimension	Include	Exclude
Population and setting	Any population and country, with low- and middle-income settings expected to dominate	None by geography
Intervention	A use case in which a blockchain or distributed-ledger component is part of delivery: payments, smart contracts, tokens, records, storage or governance	Portfolio companies with no blockchain component (UNICEF and MCV also fund AI and other technologies); token issuance for fundraising only
Comparator	Not required	None
Outcomes	Any reported operational, social or risk outcome, or none (claim-only cases are still eligible)	None
Evidence	Publicly accessible documents with case-level information: partner pages, endline reports, case studies, implementer sites, evaluations, news, code repositories and on-chain explorers	Non-public partner data (deferred to the full scope); paywalled sources that cannot be archived
Stage and time	Use cases launched from 2017 to the pilot start date, at any stage including concept-only if publicly documented	Calls for proposals with no named supported use case
Portfolio link	Publicly linked to MCV, the UNICEF Venture Fund, the Filecoin Foundation or GSR Blockchain for Good	Use cases with no public link to these four

5.2 Sampling frame

The orchestrator compiles a census of every eligible public use case in each portfolio, saved as the sampling frame (output O2). The expected starting points are:

- **MCV:** endline "Pilot Insights" posts and pilot launches on the Prosper Global Ventures Inclusive Fintech Lab and Crypto Lab pages, which publish key data and lessons from each pilot.[33][34]
- **UNICEF Venture Fund:** the Blockchain Ventures portfolio, which reports 23 blockchain solutions to date, and deployment stories.[35]
- **Filecoin Foundation:** the GSR co-funded cohort of five projects and the public Filecoin case studies.[36][37]
- **GSR Blockchain for Good:** grantees listed on the GSR Foundation website.[38][39]

The frame records the lineage ID, portfolio(s), preliminary domain, country, year, apparent deployment status and an evidence-richness flag (low, medium or high, based on the number and type of documents found).

5.3 Selection procedure

1. **Allocate by portfolio.** The target is 8 MCV, 8 UNICEF, 7 Filecoin and 7 GSR cases, for a maximum of 30. If a portfolio has fewer eligible public cases, the unused places go to the other portfolios in proportion to frame size.

[33] Prosper Global Ventures (Mercy Corps Ventures). Inclusive Fintech Lab. https://www.prosperglobalventures.com/fintech-lab

[34] Prosper Global Ventures. Crypto Lab endline posts. https://www.prosperglobalventures.com/blog/tag/Crypto+Lab+-+Endline

[35] UNICEF Office of Innovation. Blockchain Ventures. https://www.unicef.org/innovation/blockchain-ventures

[36] Filecoin Foundation (2025). Unlocking impact through infrastructure: Filecoin Foundation and the GSR Foundation work together to fund Filecoin use cases. https://fil.org/blog/unlocking-impact-through-infrastructure-filecoin-foundation-and-the-gsr-foundation-work-together-to-fund-filecoin-use-cases

[37] Filecoin. Case studies. https://www.filecoin.io/case-studies

[38] GSR Foundation. Home page (grantee listings). https://www.gsrfoundation.io/

[39] GSR Foundation. How we fund. https://www.gsrfoundation.io/how-we-fund

2. **Stratify inside each portfolio** by use-case domain and by deployment status, grouped as active (pilot or extended) versus moved on (scaled, handed over, discontinued or apparently stalled).
3. **Set coverage quotas:** at least four of the six domains in the overall sample, at least 20% (6 or more cases) stalled or discontinued, and at least one high-richness case per portfolio so the upper levels of the scale can be tested.
4. **Draw at random inside strata** using a recorded random seed, so the draw can be reproduced. A replacement is drawn the same way if a case turns out to be ineligible on full reading.
5. **Choose calibration cases.** Six of the 30 cases (at least one per portfolio, including at least one stalled case) are drawn at random as the calibration set.
6. **Log everything:** frame size, strata, seed, draws, replacements and reasons. The planned and actual samples are compared in output O3.

> **Justification.** The pilot's aim is to stress-test the taxonomy, so maximum variation matters more than statistical representativeness. Purposive strata provide that variation. Random draws within strata stop "good story" cases being chosen by hand. A failure quota counters positive publication bias and makes metric 6 (does evidence grade predict scaling?) possible.

5.4 Sample size

Thirty is the memo's ceiling and lies within the 20–30 range Campbell recommends for piloting a coding form.[40] It is enough to find taxonomy failures and estimate effort per case. It is not enough to estimate agreement precisely. With 30 cases, a kappa confidence interval can be about ±0.15 to ±0.20 wide, so agreement is reported as a first reading with its interval, and judged with that uncertainty in mind.

6. Collection plan

6.1 Search steps for each case

For each sampled use case, the orchestrator follows a fixed sequence and logs each search: the query or URL, date, result count and documents kept.

1. Partner portfolio page and any endline or impact report
2. The implementer's own website, reports and blog
3. Independent sources: evaluations, academic papers, think-tank or sector reports (for example from CGAP or GSMA), and news coverage
4. Status signals: product availability, most recent public activity, follow-on funding, hand-over or shutdown notices, code-repository activity, and on-chain activity where an address is published
5. A backward citation check on any report found

Stopping rule: stop when two consecutive steps add no new evidence-bearing document, or after a fixed cap of searches per case (set at G1). Reaching the cap is recorded in the log.

6.2 Evidence packets and archiving

Each kept document receives a source ID. Its URL, access date and a saved copy (PDF or web archive snapshot with a file hash) are stored on the restricted Drive. The orchestrator assembles an evidence packet for each case containing the verbatim document text with source IDs. The packet contains no summaries, so both coders see exactly the same material. Once coding starts, packets are frozen. Any document found later is logged and handled as a documented addition.

[40] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

> **Justification.** Web content in this field changes and disappears quickly, so archiving makes every code verifiable later. Verbatim, frozen packets make the coders' inputs identical, which is a precondition for a meaningful agreement measure.

6.3 Post-pilot status tracing

A separate step codes current status (active pilot, extended pilot, scaled, handed over, discontinued, or unknown) from the status signals collected in step 4. Confidence is high when an explicit source states the status, medium when several indirect signals agree, and low when there is a single indirect signal. Metric 4 counts cases with medium or high confidence. Status tracing is done after evidence coding, so knowing the outcome does not influence the evidence grade.

6.4 Planned versus actual

The collection plan (O4) and the collection log (O5) are kept as separate files. A deviations log (O18) records every departure from this design, including sample replacements, search caps reached, codebook changes and gate decisions, with reasons.

7. Codebook and taxonomy

Draft codebook v0.1 is summarised here, and its full field list is in Annex B. It becomes v1.0 at Gate 2, after a dry run on one non-sample case.

7.1 Matrix framework

Rows: use-case domains (subcategories)	Columns: outcome domains (examples)
1. Payments and cash transfers (humanitarian cash, anticipatory cash, remittances, merchant payments, wages)	A. Delivery efficiency (transaction cost, speed, administrative burden)
2. Credit, savings and insurance (micro-lending, savings tools, parametric insurance, guarantees)	B. Reach and access (adoption, active use, inclusion of women, unbanked people and displaced people)
3. Identity, records and data integrity (digital ID, credentials, provenance, archiving and storage)	C. Transparency, trust and accountability (traceability, leakage, fraud, user trust)
4. Supply chain and traceability (vaccines, food, commodities, tokenised goods)	D. Financial resilience and wellbeing (income, savings, coping, financial stress)
5. Climate and natural assets (carbon and environmental credits, regenerative finance, climate data)	E. Sector outcomes (health, education, climate adaptation, human rights and justice, culture)
6. Governance, funding and coordination (DAOs, community treasuries, crowdfunding, grant and donor flows)	F. Risks, harms and unintended effects (privacy, volatility, cash-out failure, exclusion, security incidents)

> **Justification.** Six by six is within Campbell's recommended range of four to six headings on each axis, which favours "a few big holes rather than many small holes".[41] The earlier scoping draft had 11 domains by 7 outcome groups. These are consolidated here, and the change is logged as codebook change v0.1-01. Outcome domain F is included because GSR's Kitemark concerns (key management, cash-out dependence) are risk outcomes that positive-results reporting tends to leave out.[42]

[41] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[42] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

7.2 Filters

Filters are characteristics that can be used to slice the map without adding axes.[43] They are: portfolio; **blockchain function** (stablecoin or payment rail, smart-contract automation, tokenisation, tamper-evident record, decentralised storage, decentralised governance, digital identity or credentials); implementation model (start-up, NGO, UN agency or country office, government, consortium); funding model; deployment status; region and country; population group; who produced the evidence; and year.

7.3 Evidence scale and crosswalk

One evidence level is coded for each outcome domain reported. A case's headline level is its highest outcome-level grade, but the matrix shows all of them.

Level	Definition	Typical evidence	Nesta	Maryland SMS	3ie map
L1 Claim	Describes what the solution does and why it should work. No data on results.	Launch post, pitch, case narrative	1	Not scoreable	Excluded
L2 Operational	Data on outputs, usage or operational performance (cost, speed, volume), with no measure of user outcomes	Transaction counts, fee comparison, on-chain data	2 (partial)	Not scoreable	Excluded
L3 Outcome, no comparison	Outcome data from participants, such as post-only or pre/post surveys or self-reports, without a comparison group	Endline survey: "64% reported higher revenue"	2	1–2	Excluded
L4 Outcome with comparison	Outcomes compared with a comparison group or a non-blockchain alternative, without strong identification	Matched comparison, difference-in-differences, comparison of delivery channels	3	3	Borderline
L5 Counterfactual impact	Randomised or strong quasi-experimental impact estimate, ideally independent	RCT, regression discontinuity, robust quasi-experiment	3–4	4–5	Included

The Nesta levels follow its five-level standards.^nesta^ The Maryland levels follow the What Works Growth version of the SMS, which treats stated effects as not scoreable.^sms^ The 3ie column shows whether a study would qualify for a 3ie effectiveness map.^wp28^

> **Justification.** The memo specifies a five-level scale.[44] An earlier six-level draft is consolidated into it, and the mapping is logged as change v0.1-02. The crosswalk lets 3ie and Campbell readers see at once how little of the evidence would qualify for a conventional effectiveness map. L2 is kept separate from L3 because GSR observes that pilots report speed and cost but rarely social outcomes, and the scale has to be able to show that.

7.4 Other judgement codes

Code	Values	Purpose
Objective achievement	Achieved / partly / not achieved / unclear, per stated objective, with a supporting quote	RQ5. Read together with the evidence level, because a claim of success at L1 is not the same as one at L4.
Blockchain added value	0 not addressed; 1 asserted; 2 descriptive comparison with a prior or alternative channel; 3 tested comparison against a non-blockchain alternative	RQ6, and an input to the Kitemark question of whether a shared ledger is needed at all[45]

[43] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[44] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[45] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

Code	Values	Purpose
Evidence producer	Implementer; partner or funder; independent; peer-reviewed	Sensitivity filter for interest-driven reporting
Claims register	Each headline claim recorded with a quote, then rated corroborated / consistent / contradicted / unverifiable	Metric 7: did headline claims hold up?
Deployment status (and confidence)	Pilot; extended pilot; scaled; handed over; discontinued; unknown	RQ3 and metric 4
Confidence (every code)	High / medium / low	Sends low-confidence codes to human review

7.5 Versioning and change log

Version numbers work as follows: v0.x are drafts; v1.0 is frozen at G2; v1.x are calibration changes approved at G3; v2.0 is the final pilot version at G4. Each change log entry records the change ID, date, field, old and new definition, what triggered it (for example a disagreement ID, a case that fit no category, or GSR feedback), the reasoning, and whether already-coded cases must be recoded. After G3, any change to a field means that field is recoded for every case. The final report includes a short narrative of how the codebook changed over time and why, as required in the outputs list.

> **Justification.** A traceable codebook history is what lets an external reviewer tell the difference between refining the method and adjusting it to fit the results. It also produces the "where the taxonomy broke down" evidence that the memo asks for in Week 3.[46]

8. Two-model coding and reliability

8.1 Independent coding

Both coders receive the same frozen evidence packet, the same codebook version and the same prompt, and run in separate sessions. They use the most deterministic settings each model allows and return a fixed structured (JSON) format. Every code must include the source ID and a verbatim quote, or "not reported". An automated check matches every quote against the packet text, and any quote that cannot be found is treated as a possible fabrication and sent to human review. Prompts, settings, model versions and dates are recorded in the model register (O8).

> **Justification.** Published comparisons show LLMs can serve as a second extractor on structured fields but are less reliable on interpretive judgements, and they recommend human verification.[47][48][49] Agreement between two models has been shown to be a useful quality signal: low agreement can point to fabrication.[50] Requiring quotes makes every code checkable.

8.2 Calibration

Both models code the six calibration cases under v1.0. The orchestrator computes agreement and sorts each disagreement by cause: codebook ambiguity, missing evidence, misreading, fabricated quote, or a genuine

[46] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[47] Jensen et al. (2025). ChatGPT-4o can serve as the second rater for data extraction in systematic reviews. https://vbn.aau.dk/ws/portalfiles/portal/766463486/Jensen_et_al._2025_._ChatGPT-4o_can_serve_as_the_second_rater_for_data_extraction_in_systematic_reviews.pdf

[48] Bianchi, J. et al. (2025). Data extractions using a large language model (Elicit) and human reviewers in randomized controlled trials. Cochrane Evidence Synthesis and Methods. https://onlinelibrary.wiley.com/doi/full/10.1002/cesm.70033

[49] Large language models with human-in-the-loop validation for systematic review data extraction (2025). arXiv. https://arxiv.org/abs/2501.11840

[50] Leveraging large language models for systematic literature review of disease spread models (2026). arXiv. https://arxiv.org/pdf/2608.26150v1

difference in judgement. Michael reviews the results at G3 and approves changes (v1.x). All 30 cases, including the six calibration cases recoded blind, are then coded under the frozen version. One further calibration round is allowed if a core field is still below 0.67.

8.3 Agreement statistics and thresholds

For each field, the pilot reports percent agreement, Cohen's kappa (weighted for the ordinal evidence level), Krippendorff's alpha (ordinal where relevant) and Gwet's AC1/AC2, with 95% bootstrap confidence intervals from resampling cases. Results are reported for all 30 cases and separately for the 24 non-calibration cases, as an out-of-sample check.

Band	Threshold	Action
Decision metric (memo)	Weighted kappa of 0.70 or more on the evidence level	Counts toward the scale-up decision rule
Good	0.80 or more	The field is used as coded
Acceptable for a pilot	0.67 to 0.79	The field is used, flagged, and revised for the full scope
Below acceptable	Under 0.67	Revised and recoded once. If still below, reported as unreliable and not used to shade the matrix.

Core fields: use-case domain, outcome domains present, evidence level for each outcome, deployment status, blockchain added value, and objective achievement. **Test-retest:** Coder A recodes five cases a few days later with identical settings to measure stability within the same model.

8.4 Human verification

Michael reviews:

- every disagreement, together with the adjudication assistant's suggested resolution;
- every L4 or L5 grade;
- every added-value code of 2 or 3;
- every low-confidence code and every failed quote match; and
- a random audit of 20% of cases (6 cases, whole records) where the models agreed.

The audit reports human–consensus agreement on core fields. The target is 90% or more. Below that, the affected field is re-examined across all cases.

> **Justification.** Two models can agree and both be wrong. The audit is the only check on correlated error, so it covers cases where the models agreed, not only disagreements. Concentrating human time on the highest grades follows the scoping memo's rule, because errors there would most distort the funding message. This mirrors rapid-review practice of dual checking a proportion and continuing with lighter checks when agreement is high.[51]

8.5 Adjudication log

Each disagreement is recorded with the case, field, both codes and quotes, the assistant's suggestion and reasoning, Michael's decision and reasoning, the error type, and any resulting codebook change ID. The final dataset keeps both original codes alongside the adjudicated value.

[51] Cochrane Training. Team considerations, study selection, risk of bias and data extraction in rapid reviews. https://training.cochrane.org/sites/training.cochrane.org/files/public/uploads/Team%20considerations,%20study%20selection,%20risk%20of%20bias%20and%20data%20extraction%20in%20rapid%20reviews.pdf

9. Analysis plan

9.1 Descriptive mapping and gap typology

Following Campbell and 3ie, the analysis is descriptive: what evidence exists, where it is concentrated, and where it is missing.[52][53] Each matrix cell is classified using a gap typology built for funding decisions:

Gap type	Definition	What it suggests for funding
Absolute gap	No use cases in the cell	Possible white space, or not a plausible use of blockchain
Outcome gap	Use cases exist, but only L1–L2 evidence	Fund outcome measurement in current grants
Counterfactual gap	L3 evidence, but nothing at L4 or above	Candidates for comparison or impact studies
Added-value gap	Outcomes evidenced, but blockchain added value of 1 or less	Fund comparisons with non-blockchain alternatives
Harms gap	No reporting on risks or harms (domain F)	Make risk reporting a grant requirement

9.2 Pilot metrics and decision rule

Metric (from the memo)	Measure in this design	Threshold
1. Agreement between the two models	Weighted kappa on the evidence level, with CI; other statistics reported alongside	0.70 or more
2. Public-source coverage	Share of cases with every core field coded as something other than “insufficient information”	80% or more
3. Taxonomy fit	Share of cases coded “none fits” on the domain field	10% or fewer
4. Post-pilot status confirmed	Share of cases with status confidence of medium or high	75% or more
5. Effort per case	Logged human time plus processing time per case, averaged	Memo threshold: 2 hours or less
6. Evidence grade and scaling	Cross-tabulation of headline level by status; Spearman correlation and Fisher’s exact test, descriptive only	First read, not pass/fail
7. Claims that held up	Share of register claims rated corroborated or consistent, compared with contradicted	Reported as a finding

Decision rule (unchanged from the memo): the full map proceeds if at least four of metrics 1–5 are met and the Week 3 method fixes need no redesign of the taxonomy or evidence scale. GSR has final authority.[54] Metric 6 is interpreted cautiously, because cases that scale tend to generate more evidence, so the direction of cause is unclear, and the sample is small.

9.3 Sensitivity analyses

- The matrix with self-reported evidence removed
- The matrix with Filecoin cases removed, to test whether data-preservation use cases fit a development-outcomes frame
- Agreement before and after codebook revision, and on calibration versus non-calibration cases
- Metric 6 using the most conservative status reading (low-confidence cases treated as “unknown”)

[52] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

[53] 3ie. Evidence gap maps (Evidence Hub). https://www.3ieimpact.org/evidence-hub/evidence-gap-maps

[54] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

10. Visual matrix

The pilot produces static images for the report and briefings. The data are stored in a structure that can be converted into an interactive map, such as EPPI-Mapper, the free tool built to support Campbell maps, at full scope.[55] A review of EGMs found that most interactive maps used a matrix design and were built with EPPI-Mapper or 3ie's software.[56]

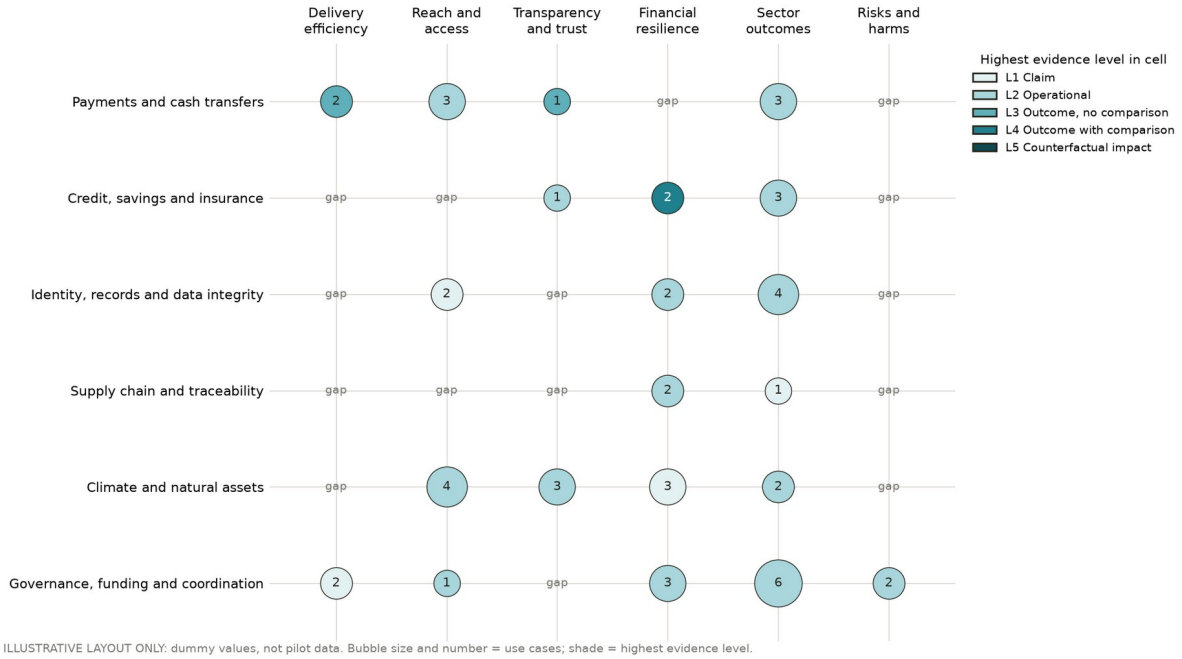


Figure 1. Illustrative layout of the core evidence matrix (V1). Values are dummy values, not pilot data.

Option	What it shows	Question it answers	Pilot
V1 Core EGM bubble matrix	Domain by outcome. Size = number of use cases; shade = highest evidence level; "gap" labels empty cells.	Where is the evidence and how strong is it?	Yes
V2 Evidence-depth bars	For each cell, a stacked bar of how many cases sit at each level	Is the evidence in a cell thin even where it is plentiful?	Yes
V3 Deployment pathway	Flow from pilot to extended, scaled, handed over or discontinued, shaded by evidence level	Are pilots inflating utility?	Yes
V4 Function by domain heatmap	Blockchain function against use-case domain	Which ledger features are used where, and is a ledger needed?	Optional
V5 Claims check	Headline claims by verification status and domain	Did reported results hold up?	Yes
V6 Funding-opportunity matrix	Cells coloured by gap type (Section 9.1), plus evaluability at full scope	Where would new grants add the most evidence?	Optional
V7 Filtered small multiples	V1 repeated by portfolio, region, population or evidence producer	Does the picture change for different slices?	Full scope

Accessibility: every visual uses labels and numbers as well as colour, uses a single-hue scale for evidence level, and is accompanied by its data table.

[55] EPPI-Centre. EPPI-Mapper. https://eppi.ioe.ac.uk/cms/er4/EPPI-Mapper

[56] Methodological characteristics of evidence and gap maps: a systematic review and evidence and gap map. Cochrane Evidence Synthesis and Methods. https://onlinelibrary.wiley.com/doi/10.1002/cesm.12096

11. Approval gates

Six gates keep human oversight at a moderate level. Michael approves every gate. GSR approves at G0 and G5 and confirms definitions at G1, as the memo provides for in Week 1.[57] Gates G1–G5 map onto the memo's four weeks.

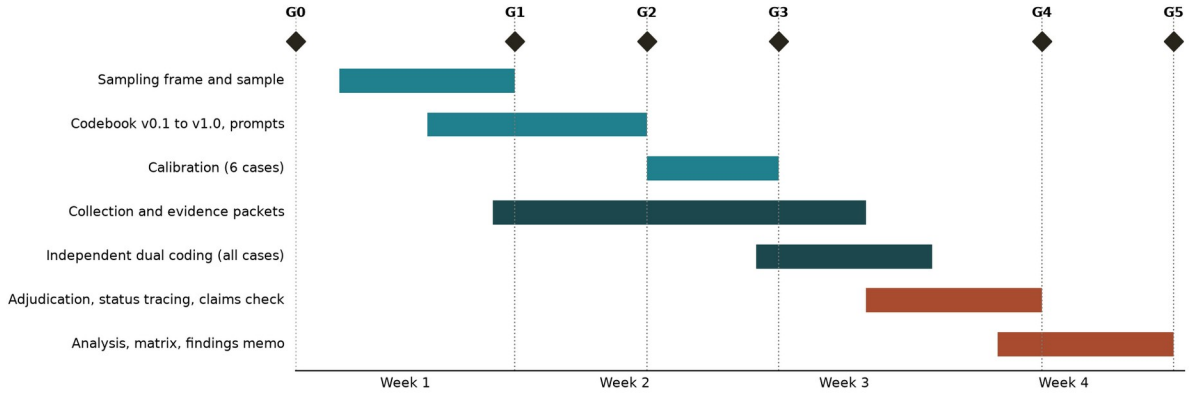


Figure 2. Indicative four-week schedule and gates. The weeks are relative, and the start date will be agreed with GSR.

Gate	Timing	Approver	Reviewed	Pass criteria	If not passed
G0 Design	Before Week 1	GSR, with Michael	This report, including the Section 2.3 changes and the Annex B codebook	GSR approval in writing	Revise and resubmit. When approved, the design is posted publicly.
G1 Sample and definitions	End of Week 1	Michael; GSR confirms definitions	Frame census, strata, seed, sample, replacements, the pilot/deployment definition, search cap	No more than 30 cases, all public; quotas met or shortfall explained	Adjust the strata or reallocate places
G2 Codebook v1.0, prompts, models	Early Week 2	Michael	Codebook, prompt pack, model register, a dry run on one non-sample case	Every field defined with an example; output format validated; quote check working	Fix and repeat the dry run
G3 Calibration	Mid Week 2	Michael	Agreement on 6 cases, error analysis, proposed v1.x changes	No core field below 0.67 after revision, or an approved fix plan	One more calibration round, then proceed with flagged fields
G4 Dataset and draft map	End of Week 3	Michael	Adjudicated dataset, agreement report, audit results, status tracing, claims register, draft visuals	All disagreements resolved; audit agreement of 90% or more; all quote failures resolved	Recode affected fields
G5 Findings and decision	End of Week 4	GSR and Metagov	Findings memo, metric results, method fixes, full-scope recommendation	Memo decision rule; GSR's judgement	Stop, or run a revised pilot

12. Outputs

ID	Output	Contents	Gate	Drive folder
O1	Pilot design report	This document, with versions	G0	01_Design

[57] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

ID	Output	Contents	Gate	Drive folder
O2	Sampling frame	Census of eligible public use cases per portfolio, with lineage IDs	G1	02_Sampling
O3	Sample plan and actual sample	Strata, quotas, seed, draws, replacements; planned versus actual	G1	02_Sampling
O4	Collection plan	Search steps, stopping rule, cap, status-signal list	G1	03_Collection
O5	Collection log (actual)	Every search with date and results; document inventory with source IDs, archived copies and hashes	G4	03_Collection
O6	Evidence packets	Frozen verbatim text for each case	G2–G4	04_Evidence_Packets
O7	Codebook and change log	Versions v0.1 to v2.0, the change log, and a narrative summary of changes and reasons	G2, G3, G4	05_Codebook
O8	Prompt library and model register	Prompts, settings, model names, versions and dates	G2	06_Prompts_and_Models
O9	Raw coder outputs	Coder A and Coder B structured outputs, plus test-retest runs	G3, G4	07_Coding_Raw
O10	Agreement report	Statistics with confidence intervals, by field and phase	G3, G4	08_Agreement_and_Adjudication
O11	Adjudication and audit log	Disagreements, suggestions, decisions, error types, audit results	G4	08_Agreement_and_Adjudication
O12	Final coded dataset	Adjudicated codes with the original codes kept, plus a data dictionary	G4	09_Final_Dataset
O13	Claims register	Headline claims and their verification status	G4	09_Final_Dataset
O14	Status tracing table	Current status, signals and confidence for each case	G4	09_Final_Dataset
O15	Visual pack	Static images V1–V3 and V5 (V4 and V6 optional), with data tables	G4	10_Analysis_and_Visuals
O16	Pilot findings memo	What the cases show, method fixes, scale recommendation. Shared only between GSR and Metagov, as the memo specifies.[58]	G5	11_Reports
O17	Methods note	ROSES-style checklist, flow diagram, AI-use disclosure	G5	11_Reports
O18	Deviations and gate log	Every departure from the design, and every gate decision	Ongoing	00_Admin_and_Approvals
O19	Full-scope protocol (if approved)	Annex A developed into a full protocol	After G5	12_Full_Scope

13. Data management, ethics and independence

- **Storage and access:** all files are kept in a restricted Google Drive folder, “Blockchain for Impact Funding Evidence Map – Pilot”, with the numbered subfolders listed in Section 12. File names follow the pattern YYYYMMDD_OutputID_Name_vX.Y. The design is made public at G0. Everything else stays restricted until GSR approves the full scope, when a project website will be built.
- **Human subjects:** the pilot uses public documents only and involves no interviews or personal data collection. Any personal data about beneficiaries that appears in documents is not extracted.

[58] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

- **Copyright:** archived copies are for internal verification only. Public outputs link to the original sources.
- **Independence:** GSR funds both this work and several of the partners. Evidence grades and conclusions are the research team's responsibility. In the full scope, partners check facts about their own use cases but do not set grades, as the memo specifies.[59]
- **Conflict of interest:** Michael Cooper has previously worked with the UNICEF Venture Fund.[60] This is declared. As a mitigation, all UNICEF cases are included in the human audit pool at twice the base rate.
- **Naming:** use cases are named, in line with the memo's public-output principle. Whether GSR is comfortable publishing weak or negative evidence on named GSR-funded cases is still an open question in the memo. The pilot findings memo is not public.

14. Risks and limitations

Risk or limitation	Effect	Mitigation
Thin public evidence	Many cases rated L1–L2; metric 2 may fail	This is a valid finding. Partner data comes in at full scope.
Positive reporting bias	Overstates success	Failure quota, evidence-producer filter, claims register
Correlated LLM errors	High agreement that is still wrong	Coders from different developers; 20% human audit of agreed codes; quote verification
Only one human	No human–human reliability measure	Disclosed. The full scope adds an external second reviewer for a subsample (Annex A).
Small sample	Wide agreement confidence intervals; metric 6 is only indicative	Report intervals; treat results as a first read
Model changes during the pilot	Results not reproducible	Pin versions, record dates, and run all coding within the pilot window
Filecoin cases fit poorly	Taxonomy-fit failures	Domain 3 and outcome E cover data preservation; sensitivity analysis
Orchestrator influence	The same model family prepares packets and suggests adjudications	Packets are verbatim; the orchestrator never codes; Michael makes every decision

[59] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

[60] Metagov, Purpose and Use Memo: Blockchain for Impact Funding Evidence Map (final draft, October 2026). Internal document held in the project Google Drive.

Annex A. Full-scope research design

This annex outlines the full Evidence Map, which will be developed into a protocol (O19) if GSR approves scale-up at G5. It keeps the pilot's framework, as revised, and brings the method closer to Campbell and 3ie practice, using partner data obtained under data-sharing agreements.

A1. Scope and tiers

Tier	Population	Coding depth	Justification
T1 Screened supply	Proposals and applications screened by partners. MCV reports insights from 800+ proposals across four funding rounds.[61]	Light: domain, function, country, population, stage	Shows what the field is attempting. Kept separate so the volume of proposals does not inflate the evidence counts.
T2 Supported use cases	All supported pilots and portfolio use cases across the four partners, plus GSR grantees	Full codebook	This is the core map
T3 Evaluability deep dives	8–15 priority use cases	Evaluability assessment, including interviews	Turns gaps into fundable evaluation opportunities
Context layer (optional)	Comparable published blockchain-for-impact evaluations from outside the four portfolios	Full codebook	Places partner evidence in the context of the wider field

A2. Data-sharing agreements

Agreements with each partner will cover: the data in scope (application records, monitoring data, endline datasets and reports); de-identification and aggregation rules; permitted uses; what may be published and in what form (including use-case naming); the partner's factual-review rights and timelines; storage, security and retention; and attribution. T1 data will normally be shared in aggregated or anonymised form. The minimum terms for data management, security and privacy are in A10, and the itemised data request is in A11.

A3. Search and screening

- A census of the partner portfolios using shared data, plus the public sources used in the pilot
- A documented systematic grey-literature search, with search strings, across development, humanitarian and academic sources (for example 3ie's evidence portal, ReliefWeb, Google Scholar, SSRN and sector bodies), together with a public call for evidence and backward and forward citation tracking, in line with 3ie practice[62]
- Dual independent screening by two models, with human resolution of every conflict and a random 10% human check of agreed exclusions, reported in a PRISMA/ROSES flow diagram[63]

A4. Coding and reliability at scale

- Codebook v2.0 from the pilot is frozen as v3.0 after a further calibration round on 20 new cases.
- Agreement is monitored continuously in batches of about 50 cases, with alerts if any core field falls below 0.67 and checks for drift between batches.
- Models are pinned to fixed versions. Any model change triggers revalidation on a reference set of 20 cases.
- Human audit: all L4–L5 grades and added-value codes of 2 or 3; a stratified random 10% of agreed codes; and, where funding allows, an external second human reviewer on a 10% subsample to give a human–human benchmark.

[61] Prosper Global Ventures (Mercy Corps Ventures). Inclusive Fintech Lab. https://www.prosperglobalventures.com/fintech-lab

[62] 3ie (2025). Private sector engagement evidence gap map protocol. https://3ieimpact.org/sites/default/files/2025-05/PSE-EGM-Protocol.pdf

[63] ROSES for systematic map reports. https://www.roses-reporting.com/systematic-map-reports

A5. Critical appraisal

L4–L5 studies receive a light risk-of-bias assessment suited to their design (randomised versus non-randomised). L2–L3 sources receive a short reporting-quality checklist covering data source, sample, measurement period, attrition and independence. Appraisal results are shown as a map filter.

A6. Evaluability layer

Following Davies (2013), each T2 case is pre-scored on evaluability in principle (design and theory of change), in practice (data and systems) and in utility (stakeholder demand).[64] T3 cases are verified through interviews and document review. Each case is classified as evaluable now, evaluable with fixes, monitor only, or not worth evaluating, which feeds visual V6.

A7. Stakeholders and review

An advisory group including GSR, the four partners and an external evidence-synthesis methodologist reviews the framework and the draft map, consistent with 3ie's use of advisory groups at the start and end of mapping.[65] Partners check facts about their own use cases before publication. The research team keeps responsibility for grades.

Participation is a primary purpose of the full scope, so that the map and its products are useful for decisions and joint learning by each partner. Each partner is invited to be a full partner: a seat on the advisory group, co-design of the codebook and case list, named partner credit and, by agreement, co-authorship. Partners receive briefings and updates at a frequency they choose. Participatory milestones are: (1) kickoff and co-design; (2) validation of the case list; (3) preliminary coding results; (4) preliminary results, including partner validation before publication; (5) a workshop on using the final products; (6) a post-launch learning session; and (7) any other point of collaboration a partner requests. Validation means partners correct facts and comment on interpretation; results are published only after this review, and the research team keeps responsibility for evidence grades.

A8. Outputs, website and updating

- A project website with an interactive map (EPPI-Mapper or equivalent), filters (V7) and case records[66]
- A report written for non-specialists, a plain-language summary, a technical methods annex and a ROSES checklist[67]
- An open dataset, the codebook and the prompts, plus general guidance on grantee reporting that uses the map's fields
- A living-map process so grantee reports can be added, with at least an annual update as Campbell recommends[68]

A9. Full-scope gates

Gate	Decision
F0	Full protocol approved by GSR, then posted publicly. Formal registration (for example on OSF or with Campbell) to be considered at that point.
F1	Data-sharing agreements signed and data received

[64] Davies, R. (2013). Planning evaluability assessments: a synthesis of the literature with recommendations. DFID Working Paper 40. https://www.gov.uk/government/publications/planning-evaluability-assessments

[65] 3ie. Evidence gap maps (Evidence Hub). https://www.3ieimpact.org/evidence-hub/evidence-gap-maps

[66] EPPI-Centre. EPPI-Mapper. https://eppi.ioe.ac.uk/cms/er4/EPPI-Mapper

[67] Campbell Collaboration (2020). Editorial: Evidence and gap maps. https://www.campbellcollaboration.org/2020/03/editorial-journal-issue1-2020/

[68] White, H. et al. (2020). Guidance for producing a Campbell evidence and gap map. Campbell Systematic Reviews 16(4): e1125. https://onlinelibrary.wiley.com/doi/abs/10.1002/cl2.1125

Gate	Decision
F2	Codebook v3.0 and screening calibration approved
F3	T2 coding, audit and agreement report approved
F4	Draft map and report after partner factual review and advisory group review
F5	Publication and website launch

An indicative 12–18 weeks for the full map was proposed at the scoping stage. This will be re-estimated from the pilot's effort-per-case results (metric 5).

A10. Data management, security and privacy

This section sets the data-management terms for the full scope. It applies to all non-public partner data, which the pilot does not use, and to the same standard for every partner (Mercy Corps Ventures, the UNICEF Venture Fund, the Filecoin Foundation and GSR). Each data-sharing agreement (A2) adopts these terms as a minimum; where a partner's own policy is stricter, the stricter rule applies. The terms are designed to meet:

- **UNICEF Policy on Personal Data Protection (2020):** its eight commitments that processing be justified, for defined purposes, limited to what is necessary, accurate, secure and confidential, limited in time, transparent to data subjects, and protected on transfer; classification of personal data by sensitivity; and logged access.
- **UNICEF Procedure on Ethical Standards in Research, Evaluation, Data Collection and Analysis (2021):** ethical review for work involving human subjects, sensitive secondary data or novel technologies such as AI-assisted processing; informed consent; privacy and confidentiality; and evaluator independence.
- **Mercy Corps Responsible Data Policy:** a privacy impact assessment, a sensitive information assessment for any sensitive data, and encryption of sensitive data at every stage.
- **Good research practice:** the Five Safes framework; J-PAL data security and de-identification guidance; the IASC and OCHA data responsibility guidance; the 3ie Research Transparency Policy; and Campbell's open-access and FAIR data expectations.

A10.1 Roles and governance

- **Controller and processor:** each partner remains the controller of the data it shares. Metagov processes it only on the partner's documented instructions, as set out in the agreement.
- **Data custodian:** Michael Cooper (Research Director, Metagov) is accountable for compliance. Each partner names a data contact.
- **GSR access:** GSR receives only aggregated or partner-approved outputs, not raw partner data, unless the partner agrees in writing.
- **Records:** a data register lists every dataset, its owner, class, storage location, access list, permitted use and deletion date. Decisions and incidents are logged in O18. Compliance is reviewed at every approval point and participatory milestone.

A10.2 Data classification

Every item is classed before transfer. The class sets the handling rules.

Class	Examples	Storage and handling	Access
Public	Published reports, websites, public case studies	Any project storage; cited with source links	Anyone
Internal	Portfolio lists, status updates, aggregated screening counts	Restricted Shared Drive	Research team and the partner
Confidential	Investment memos, agreements, monitoring and endline reports, de-identified datasets	Restricted Shared Drive, partner folder; no download for viewers	Named users and the partner

Class	Examples	Storage and handling	Access
Sensitive (restricted)	Personal data such as interview contacts, recordings and transcripts; any data on children, displaced people, health or finances; wallet addresses linked to individuals; security incidents	Restricted Shared Drive plus file-level AES-256 encryption at all stages; held apart from analysis files	Data custodian and named interviewers only

A10.3 Purpose and legitimate basis

- **Basis:** partner data is processed under the data-sharing agreement; interviews are processed on the basis of informed consent.
- **Purpose limitation:** data is used only for the Evidence Map purposes named in the agreement. It is not sold, reused for other projects or used to train AI models. Any new use needs the partner's written agreement.

A10.4 Assessments and ethics review

- **Impact assessment:** one combined assessment, meeting both UNICEF's data protection impact assessment and Mercy Corps' privacy impact and sensitive information assessments, is completed before any personal or sensitive data is received. It is shared with the partners.
- **Ethics review:** the T3 interviews and the AI-assisted processing of partner documents go to independent ethics review before they start: an accredited institutional review board, a panel of at least three independent experts, or UNICEF's ethics review arrangements where UNICEF prefers.
- **Scope limits:** the study does not collect data from children or programme beneficiaries.

A10.5 Minimisation and de-identification

- **Minimum necessary:** the study requests only the items listed in A11. Partners remove personal data before transfer, and T1 screening data is shared aggregated or with applicant contact details removed.
- **Blockchain identifiers:** on-chain wallet addresses and transaction hashes linked to individuals are treated as personal data. They are not requested and never published.
- **Unexpected personal data:** if personal data arrives, it is isolated, the partner is told, and the data is deleted or de-identified within five working days. The event is logged.
- **Separate identifiers:** identifying information needed for follow-up is kept in a separate encrypted file. Analysis files use random study IDs; lineage IDs are never derived from names.
- **Disclosure control:** before release, outputs are checked for small cells and combinations of indirect identifiers (for example country, year and programme size) that could identify a person or a small organisation, with both personal and group privacy considered.

A10.6 Consent and the rights of data subjects

- **Informed consent:** interviewees are told the purpose, who will see their information, how long it is kept, whether it may be reused, the limits of confidentiality and their right to withdraw. Consent is recorded.
- **Requests:** requests to see, correct or delete personal data are answered within 30 days, in coordination with the partner.

A10.7 Safe people and access

- **Named users only:** anyone given access signs a confidentiality undertaking and completes research ethics and data-protection training first.
- **Least privilege:** each partner sees only its own cases and data. Access is reviewed at every approval point and removed when a role ends or the project closes.

A10.8 Storage, security and transfer

- **Restricted storage:** partner data is held in a dedicated Google Shared Drive in Metagov's Google Workspace, not a personal account. There is one folder per partner, link sharing is off, external sharing is limited to named

partner accounts, two-step verification is required, and download, print and copy are turned off for viewers. Drive audit logs are reviewed at each approval point where available.

- **Encryption:** data is encrypted in transit and at rest by the platform. Sensitive files are also encrypted at file level, with passwords sent by a separate channel. Working devices use full-disk encryption and screen locks and keep no local copies after a working session.
- **GitHub:** the private workspace repository holds only code, the codebook, prompts, the model register and de-identified analysis tables. Raw partner files and personal data never go into any GitHub repository. The public repository receives only approved outputs.
- **Transfer:** data moves only by Drive sharing to named accounts or the partner's own secure transfer tool, never as email attachments or in chat. Onward transfer to anyone else needs the partner's written agreement.
- **Service providers:** the agreement lists every service that touches partner data (Google Workspace and the AI providers below). No other provider is used without notice to the partner.
- **Backups:** Drive version history plus one encrypted backup held under the same access rules.

A10.9 AI processing

- **Approved services only:** partner documents are processed only through paid API or enterprise tiers whose terms exclude training on customer inputs (for example the OpenAI API and Gemini API Paid Services), with zero data retention requested where offered. Consumer chat apps are never used.
- **Safeguards:** personal data is removed before model processing, each partner consents in writing to AI-assisted coding, a risk assessment of automated processing is part of the impact assessment, and the model register records which provider processed which dataset. A human decides every grade.

A10.10 Validation, release and open licence

- **Release checklist:** nothing leaves restricted storage until five checks are complete: partner validation of its own cases (normally ten working days), a disclosure check, agreement on case naming, confirmation that no confidential material is included without written approval, and GSR approval at the launch approval point.
- **Open licence:** all research materials are published for public use: the protocol, reports, codebook, evidence matrix, coded dataset, adjudication logs and visuals under Creative Commons Attribution 4.0 (CC BY 4.0), and code and AI prompts under the MIT licence. This follows Campbell's CC BY 4.0 open-access practice so that others can assess, replicate and reuse the methods.
- **Limits of the licence:** raw partner data, personal data, interview recordings and transcripts, and confidential partner documents are not licensed or published. As the 3ie Research Transparency Policy requires, the public record describes what is withheld, why, and how qualified researchers can request access through the partner.
- **Data availability:** the published dataset carries a data dictionary and a data availability statement, and is deposited in a citable repository (for example Zenodo or the Harvard Dataverse) in line with FAIR principles.

A10.11 Retention, deletion and incidents

- **Retention:** raw partner data is kept until 12 months after the final report, or for the period the agreement sets, then deleted or returned to the partner with written confirmation. Interview recordings are deleted once transcripts are checked. The de-identified coded dataset is kept and published.
- **Incident response:** a suspected breach is contained immediately. The partner and GSR are told within 72 hours, or sooner if the partner's procedure requires it (for example UNICEF's Personal Data Breach Procedure). The incident is logged in O18, affected people are informed where appropriate, and the controls are reviewed.
- **Concerns:** partners and interviewees can raise ethics or data concerns with the data custodian or, in confidence, with GSR.

A11. Itemised data request

This is the full list of what the full scope asks each partner for, under the data-sharing agreement. Partners share only what they hold and agree to share. Classes follow A10.2; the fields named are those in the codebook (Annex B).

Item	Tier and purpose	Form	Class
1. List of blockchain-related applications or proposals screened, by funding round	T1: volume, domains, countries and stage of everything considered	Aggregated counts, or a list with applicant contact details removed	Internal
2. List of supported or invested use cases	T2: the census frame; IDs, primary portfolio, country, year launched, funding round and amount band	Spreadsheet	Internal
3. Application, proposal or investment memo for each supported case	T2: problem addressed, stated objectives (up to 3), blockchain function, implementation model, delivery partners, funding model and target population	Documents	Confidential
4. Grant or investment agreement milestones and KPIs	T2: what was promised, for objective-achievement coding	Extract or summary	Confidential
5. Progress and monitoring reports or dashboards	T2: outputs, usage and reach (adoption, active users, disaggregation by sex and displacement status), for evidence levels L2–L3	Reports; aggregated figures only	Confidential
6. Endline, close-out and evaluation reports, with de-identified datasets where they exist	T2: outcomes, comparison groups and evidence levels L3–L5, plus critical appraisal	Reports; de-identified data	Confidential
7. Post-funding status of each case	T2: deployment status (scaled, handed over, discontinued) and reasons, follow-on funding	Short update form	Internal
8. Any comparison with non-blockchain alternatives	T2: blockchain added-value code	Documents	Confidential
9. Known risks, harms or incidents	T2: outcome domain F (privacy, security, volatility, cash-out failure, exclusion)	Summary; no personal details	Sensitive
10. Contacts for priority cases	T3: arranging interviews with staff and implementers	Name, role and work email only	Sensitive
11. Interviews (45–60 minutes) on 8–15 priority cases	T3: verifying status, evaluability in principle, in practice and in utility	Consented recordings and transcripts	Sensitive
12. The partner's data-protection requirements and naming preferences	All tiers: terms of the agreement and publication	Policy documents	Internal

Not requested: beneficiary-level records; data on children; individual wallet addresses or transaction data linked to people; bank or mobile-money account details; health records; and applicants' personal contact details.

Annex B. Draft codebook v0.1 field list

Section	Field	Type and values
Identification	Use-case ID; lineage ID; solution ID; name; primary portfolio; all funders	Text and IDs
Identification	Country; region; year launched	List; ISO year
Description	Problem addressed; stated objectives (up to 3)	Text with quotes
Description	Use-case domain and subcategory	Single choice (1–6) plus “none fits” and a note
Description	Blockchain function(s)	Multiple choice (7 options plus other)
Description	Implementation model; delivery partners; funding model	Single choice plus text
Description	Population group(s); reported reach	Multiple choice; number with quote
Outcomes	Outcome domains reported (A–F)	Multiple choice; “not reported” allowed
Outcomes	For each domain: evidence level L1–L5; key result; quote; source ID	Ordinal, text and ID
Outcomes	Evidence producer for each source	Implementer / partner or funder / independent / peer-reviewed
Judgements	Objective achievement for each objective	Achieved / partly / not / unclear
Judgements	Blockchain added value	0–3
Judgements	Risks or harms reported	Yes/no, with a list
Status	Deployment status as stated in the sources	Six categories
Status	Current traced status; signals; confidence	Six categories; list; high/medium/low
Claims	Headline claims (up to 5): text, quote, verification	Corroborated / consistent / contradicted / unverifiable
Quality control	Confidence for each code; “insufficient information” flag; coder ID; codebook version	High/medium/low; yes/no; text

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